Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 28)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Do not write outside the box

Answer all questions in the spaces provided.

0 1 Table 1 shows three specialist solid wastes.

0 1 . 1 Complete Table 1 to show the main hazard of each waste and the disposal method used.

[6 marks]

Table 1

Waste	Main hazard	Disposal method	
Asbestos			
Cyanide compounds			
High-level radioactive waste			

0 1 . 2 Explain why encapsulated asbestos in cement roof panels is not considered dangerous while it is intact.

[1 mark]

0 1 . 3 Describe the process of encapsulation used for intermediate-level radioactive waste and wastes containing heavy metals.

[2 marks]

Do not write outside the box

Do not write outside the box

.....

.....

.....

.....

0 1 . 4 Explain why vitrification is preferred to encapsulation in cement for high-level waste.

[1 mark]

.....

.....

10

0 2 A student determined the water and organic matter content of a soil sample. Table 1 shows the results.

Table 1

Measurement	Mass / g
Empty crucible	25.0
Crucible + fresh soil	65.0
Crucible + soil after drying at 105 °C to constant mass	55.0
Crucible + soil after heating in a furnace at 550 °C	52.6

0 2 . 1 Calculate the percentage water content of the fresh soil.

[2 marks]

Answer ..... %

0 2 . 2 Calculate the percentage organic matter content of the dry soil.

[2 marks]

Do not write outside the box

Answer ____ %

0 2 . 3 Explain why the soil is dried to constant mass before the organic matter is burnt off.

[2 marks]

.....

.....

.....

.....

0 2 . 4 Explain how the organic matter content affects the fertility and erosion risk of this soil.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 2 . 5 Suggest **one** reason why two students measuring the same field might get different organic matter results.

[1 mark]

.....

.....

10

0 3 . 1 Describe **three** effects of noise on human health other than hearing damage.

[3 marks]

1 .....

.....

2 .....

.....

3 .....

.....

0 3 . 2 Explain how noise can damage non-living objects.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 3 . 3 Explain how noise can reduce the breeding success of birds.

[2 marks]

.....

.....

.....

.....

0 3 . 4 Explain why military sonar is a threat to cetaceans.

[2 marks]

.....

.....

.....

.....

10

Do not write outside the box

0 4 . 1 Give **three** sources of oil pollution of the sea other than tanker accidents.

[3 marks]

1 .....

.....

2 .....

.....

3 .....

.....

0 4 . 2 Explain how an oil spill at sea can kill seabirds.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 4 . 3 Explain **two** ways in which an oil slick can harm organisms that live in the water beneath it.

[2 marks]

.....

.....

.....

.....

0 4 . 4 Explain why oil pollution is most serious when it occurs inshore or in an enclosed body of water.

[2 marks]

.....

.....

.....

.....

Do not write outside the box

10

0 5 Table 1 shows the average annual radiation dose received by a person in the UK from different sources.

Table 1

Source	Percentage of average dose / %
Radon gas from the ground	50
Medical (X-rays, CT scans)	16
Gamma rays from rocks / buildings	13
Internal (food and drink)	9.5
Cosmic radiation	10
Occupational, fallout, nuclear discharges, consumer products	1.5

0 5 . 1 Explain why radon concentrations are high inside some houses in Devon and Cornwall.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 5 . 2 Use Table 1 to calculate the percentage of the average dose that comes from artificial (human-made) sources.

[1 mark]

Answer ..... %

0 5 . 3 Suggest **two** reasons why the dose received by an individual may differ greatly from the average.

[2 marks]

Do not write outside the box

1 .....

.....

2 .....

.....

0 5 . 4 Explain why a risk:benefit analysis is needed for the use of X-rays in medicine, and give **two** reasons why the risks of ionising radiation are difficult to assess.

[4 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

10

0 6 . 1 Which of the following water sources is usually the most convenient for large communities? Tick one box.

[1 mark]

A Desalinated seawater ☐

B River water ☐

C Rainwater catchment ☐

D Newly discovered aquifers ☐

0 6 . 2 Explain why the quality requirements for water differ between uses, using **two** named uses.

[2 marks]

.....

.....

.....

.....

Do not write outside the box

0 6 . 3 Explain why inter-basin transfers may be used and give **one** disadvantage.

[2 marks]

.....

.....

.....

.....

5

0 7 . 1 Which of the following properties would make a pesticide **least** likely to biomagnify? Tick one box.

[1 mark]

A High liposolubility and high persistence ☐

B Low liposolubility and low persistence ☐

C High liposolubility and low persistence ☐

D Low liposolubility and high persistence ☐

0 7 . 2 Explain the difference between **bioaccumulation** and **biomagnification**.

[2 marks]

.....

.....

.....

.....

0 7 . 3 Explain why pesticides are described as an unusual type of pollutant.

[2 marks]

.....

.....

.....

.....

5

Do not write outside the box

0 8 Table 1 shows three investigations carried out by a class.

Table 1

Investigation	Data collected
A	Mean height of bluebells in a coppiced and an uncoppiced wood (both normally distributed)
B	Number of limpets in each of five zones on a rocky shore, compared with the numbers expected if they were evenly spread
C	Soil moisture and the percentage cover of a moss, measured at 14 points along a transect

0 8 . 1 Name the most suitable statistical test for each investigation in Table 1.

[3 marks]

0 8 . 2 Explain what it means if a test gives a probability of $p = 0.02$.

[2 marks]

0 8 . 3 Explain why the calculated value from a test must be compared with a critical value, and what else is needed to find that critical value.

[3 marks]

Do not write outside the box

Do not write outside the box

0 8 . 4 Explain what the **standard deviation** of each set of bluebell heights tells the student, and how it is shown on a graph of the means.

[3 marks]

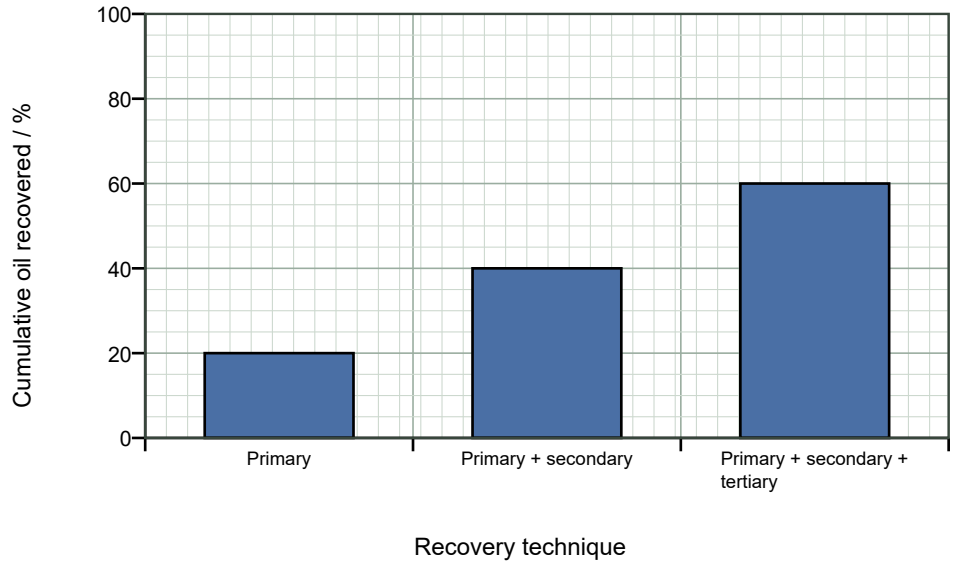
0 8 . 5 The class found a significant correlation in investigation C. Explain **four** reasons why this does not show that soil moisture causes the moss to grow.

[4 marks]

15

0 9 Figure 1 shows the percentage of the oil in a reservoir that can be recovered using different recovery techniques.

Figure 1



0 9 . 1 Describe how primary oil recovery works and explain why it recovers only about 20% of the oil.

[2 marks]

.....

.....

.....

.....

0 9 . 2 Describe secondary recovery and explain how it can be linked to carbon capture and storage.

[2 marks]

.....

.....

.....

.....

0 9 . 3 Describe **three** tertiary (enhanced) recovery techniques.

[3 marks]

Do not write outside the box

1 .....

.....

2 .....

.....

3 .....

.....

0 9 . 4 Explain **three** advantages of directional drilling.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

10

1 0 . 1 Explain why the development of energy storage technologies is essential if renewable energy resources are to provide most of a country's electricity.

[9 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

Do not write outside the box

Do not write outside the box

Extra space

.....

.....

.....

.....

1 0 . 2 State what is meant by **peak shaving**.

[1 mark]

.....

.....

10

Do not write outside the box

Write an essay on one of the following topics.

1 1 . 1 Mineral resources are non-renewable. Discuss how an understanding of geological processes and the development of new technologies can extend the period for which society can rely on mineral resources.

[25 marks]

OR

1 1 . 2 Discuss how knowledge of the properties of pollutants can be used to reduce the severity of their impact on the environment.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

Do not write outside the box

Do not write outside the box

Do not write outside the box

Do not write outside the box

END OF QUESTIONS